Exercises with hints and tips

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Now it's time to put into practice the concepts you have learnt so far in this module. Attempt the following exercises. If you get stuck, you can refer to the hints and tips at the bottom of the page.

Please note that these exercises are optional, further practice – but we strongly recommend that you engage with them in order to test your knowledge and see where you might need to do additional study.

1. A bowl contains 10 red balls and 10 blue balls. A woman selects balls at random without looking at them.
    a. How many balls must she select to be sure of having at least three balls of the same colour?
    b. How many balls must she select to be sure of having at least three blue balls?
2. Show that if seven distinct integers are selected from the first 10 positive integers, there must be at least two pairs of these integers with the sum of 11.
3. Show that if there are 30 students in a class, then at least two have last names that begin with the same letter.
4. Suppose that there are nine students in a class at a small college.
    a. Show that the class must have at least five male students or at least five female students.
    b. Show that the class must have at least three male students or at least seven female students.

Hints and tips for each question:

1. Use the extended version of the Pigeonhole Principle. Each colour is a place and the balls are pigeons.
2. Each pair whose sum is 11 is a place and those 10 numbers are pigeons.
3. We have 26 places (each letter is a place) and we have 30 pigeons.
4. There are two places: males and females. Students are our pigeons.

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